

Exercise 3

Installation and configuration of ffmpeg and ffprobe:

My first attempt was to just download static build from official website:

<https://www.ffmpeg.org/download.html#build-mac>

But then I was running into issues – related to mac security and path set.

To troubleshoot I came across this article:

<https://phoenixnap.com/kb/ffmpeg-mac> Title: Installing FFmpeg on mac {3 Methods Explained} Author: Marko Aleksic and. Accessed: Aug 20 2024

This article explains how to add static build to path but also shows other method using brew. Since I was familiar with brew and had it installed on my machine I decided to try it so I used following command:

- brew install ffmpeg

and that worked perfectly. It installed both ffprobe and ffmpeg and I was able to call them from terminal and Jupiter on my visual studio code without any issues.

Application analysis:

Application has 2 functions. One is focused on analyzing single video, the other is processing entire folder of files and calls analyzeVideo on every single file within that folder.

analyzeVideo

takes two parameters: file to be analyzed, and path for new file if necessary

- set variables for specified aspects
- get info from file using ffprobe
- make variables from info from file with same aspects as specified
- create string that will be ffprobe command if any aspect does not fit specification and report string to store information which parts were not fitting
- compare each aspect with specification and add to chain to address the issue, as well add note to report string
- if any aspect needed adjustment create new file and return string report

processFolder –

takes two parameters: folder to process, folder where to save new files if needed

- make report string
- loops over every file in folder and calls analyzeVideo on it, and where to save new
- adds reports from each file to report string
- creates report file and puts all the files report in it

Terms Description:

Video format is a designed way to store video and audio data in one file that follows certain standards. Video and audio data has to have structure to be stored or play, therefore there are codecs that are used to encode and decode data in certain organized way. Frame rate relates to video, and it is a measurement of how many images per second is shown. Aspect ratio is ratio of width to height. Resolution is the number of pixels in a frame – the more pixels the more detailed image is. Video bit rate is measurement that counts how many bits is needed to store video data per second of the video in case of video resolution and number of rates per seconds impacts that amount, in case of audio bit rate it is same concept but for audio and effecting factors are sample rate, bit depth and number of channels. Audio channels is the number of individual audio streams in one file, this allows for different levels of spatial experience but also effects size of stored audio data.